A Mathematical Foundation of Patent US 9,043,698 B2 - Draft

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The goal of this paper is to study and prove the validity and soundness of the logical $encoding$ introduced in patent US 9,043,698 B2. Given the goal, the motivation is to establish $logic$ diagrams using principles and techniques from broader areas of $mathematics$ and $computer\ science$ and to apply these findings in different areas of computation.

1 Introduction

$$\mathbf{G} = \{F, T\}$$

To study $binary\ logic$, describe $unary$ operators on the first group. The set **G** = {F, T} interpreted as $fallacy\text{-}tautology$ (or as $first\text{-}last$ which will be helpful in later $compounding$ of $operations$.) The group is $\mathcal{G}$ = {**G**, •} . This journey will continue on a line of reasoning to construct a $logical\ map$ from this $Boolean\ algebra$.

$$\mathcal{G} = \{\mathbf{G}, \bullet\}$$

There is also a $non\text{-}permutative$ (not $bijective$) $mappings$ and this non-standard notation, for example, $((A)B)$. This is "non-permutative" because A and B are both mapped to A. It is not $one\text{-}to\text{-}one$ nor $onto$ so it cannot be a $permutation$ but the digression will be productive.

The above mappings are defined such that the value of $\varphi(A)$ is the permutation applied to A. The $fallacy$ unary operator maps (F) to itself and maps T to (F).

$$A \bullet ((F)T) \mapsto F$$

$$Fallacy : \varphi(A) \equiv A \bullet ((F)T)$$

A	F	F
	T	F

Figure 1: Non-Permutative Mapping of $Fallacy$

$Tautology$ maps F to (T), and (T) to T.

$$A \bullet (F(T)) \mapsto T$$

$$Tautology : \varphi(A) \equiv A \bullet (F(T))$$

A	F	T
	T	T

Figure 2: Non-Permutative Mapping of $Tautology$

$Fallacy$ and $tautology$ are not $surjective$ so certainly not $bijective$ and this means they are $non\text{-}permutative$, and are two of four $combinations$ that are possible for unary operators.

Introducing two permutative operations as unary operators: $(F)(T)$ and (FT).

$$Is : \varphi(A) = A \bullet (F)(T)$$

A	F	F
	T	T

Figure 3: Permutation Is

$$Not : \varphi(A) = A \bullet (FT)$$

A	F	T
	T	F

Figure 4: Permutation Not

The above permutations are $bijective$ under $\mathcal{G}$: $\{F \mapsto T, T \mapsto F\}$. The $bijective$ permutation operates under $\mathcal{G}$: $\{F \mapsto F, T \mapsto T\}$.

Introducing two permutative operations as $unary$ operators: σ = $(F)(T)$ and τ = (FT).

$$\tau = (FT)$$
$$\tau(A) = A\bullet(FT)$$

$$\sigma = (F)(T)$$
$$\sigma(A) = A\bullet(F)(T)$$

A is mapped by σ to itself.

The above two permutations are the only permutations under $\mathcal{G}$ because those are the only $bijective$ operations. The other two possible mappings are neither $bijective$ nor $surjective$ and so are not $permutative$ under $\mathcal{G}$.

Under group $\mathcal{G}$, we can proceed to introduce the operation $\bullet$ and compound the elements of $\mathcal{G}$ as a group of $permutations$.

$$\mathbf{P} = \{F \bullet F, F \bullet T, T \bullet F, T \bullet T\}$$

$F \bullet F$		B	
		F	T
A	F	F	F
	T	F	F

$F \bullet T$		B	
		F	T
A	F	F	F
	T	T	T

$T \bullet F$		B	
		F	T
A	F	T	T
	T	F	F

$T \bullet T$		B	
		F	T
A	F	T	T
	T	T	T

Introducing two permutative operations as $unary$ operators: $\sigma = (F)(T)$ and $\tau = (FT)$. This is a new permutation group $\mathcal{P}$.

$$\mathcal{P} = \{\sigma, \tau, \bullet\}$$

$$\tau(F) = T$$
$$\tau(T) = F$$

Two observations:

$$\tau^2(F) = F$$
$$\tau^2(T) = T$$

Then, the other $transposition$ is,

$$\sigma = (FT)$$
$$\sigma(A) = A\bullet(FT)$$

σ maps A to A.

$$\sigma(F) = F$$
$$\sigma(T) = T$$

σ takes F to F, and T to T. In other words, $\sigma(A) = A$.

$$A \bullet (F)(T) = A$$

However, τ takes F to T, and takes T to F which in essence is $\varphi(A) \neq A$, or alternatively $\varphi(A)$ $is\ not\ equal\ A$, $\varphi(A)$ $is\ equal\ not\ A$, and $\varphi(A) = not(A)$.

$$A \bullet (FT) = not(A)$$

Two observations:

$$\tau^2(F) = \sigma(F)$$
$$\tau^2(T) = \sigma(T)$$

In other words,

$$(FT)(FT) = (F)(T)$$

$$\mathbf{Q} = \{\sigma \bullet \sigma, \sigma \bullet \tau, \tau \bullet \sigma, \tau \bullet \sigma\}$$

$\sigma \bullet \sigma$		B	
		F	T
A	F	F	T
	T	F	T

$\sigma \bullet \tau$		B	
		F	T
A	F	F	T
	T	T	F

$\tau \bullet \sigma$		B	
		F	T
A	F	T	F
	T	F	T

$\tau \bullet \tau$		B	
		F	T
A	F	T	F
	T	T	F

$$\mathbf{R} = \{F \bullet \sigma, T \bullet \sigma, F \bullet \tau, T \bullet \tau\}$$

$F \bullet \sigma$		B	
		F	T
A	F	F	F
	T	F	T

$T \bullet \sigma$		B	
		F	T
A	F	T	T
	T	F	T

$F \bullet \tau$		B	
		F	T
A	F	F	F
	T	T	F

$T \bullet \tau$		B	
		F	T
A	F	T	T
	T	T	F

$$\mathbf{S} = \{\sigma \bullet F, \sigma \bullet T, \tau \bullet F, \tau \bullet T\}$$

$\sigma \bullet F$		B	
		F	T
A	F	F	T
	T	F	F

$\sigma \bullet T$		B	
		F	T
A	F	F	T
	T	T	T

$\tau \bullet F$		B	
		F	T
A	F	T	F
	T	F	F

$\tau \bullet T$		B	
		F	T
A	F	T	F
	T	T	T

This paper introduces an approach to establishing Boolean logic from a group $\mathcal{G}$ = $\{\mathbf{G}, \bullet\}$ where $\mathbf{G}$ = {T, F}. The description herein begins with $fallacy$ and $tautology$, proceeding to the $unary$ and $binary$ operators, the construction continuing to be made in a group theoretic way so as to create a richer understanding of Boolean logic and structure.

Expanding upon this endeavour, $left\text{-}identity \bullet right\text{-}draw$, and the complementary $left\text{-}draw \bullet right\text{-}identity$ are introduced.

Define the $A\ draw$ on the permutation $\mathcal{P}_\square$ as $A \bullet \mathcal{P}_\square$ so that the following are equivalent.

$$\varphi(A) = A \bullet ((F)T)$$
$$\mathcal{P}_{\square} = ((F)T)$$
$$\varphi(A) = A \bullet \mathcal{P}_{\square} = \mathcal{P}_A$$

where

$$\mathcal{P}_{\square} = \begin{cases} P_F = F \\ P_T = F \end{cases}$$

$First$ binary operator to consider is as good as any, namely, $A \bullet B = B$. This is $left\text{-}identity$ • $right\text{-}draw$ because whatever value of A, the value of B is the $draw$ because A is the $left\text{-}hand\ identity$.

A and B are labels of primitive values F, $first$, and T, $last$, where T is the capitalized last character of 'last'. The name of the binary operation is drawn with the symbol •, written $A \bullet B$ and is read as '$A\ draws\ B$'. The $left\text{-}identity \bullet right\text{-}draw$ is the result of the left operand drawing the right whatever the value of A (F or T).

A $draws$ B.

$$A \bullet B = B$$

$Draw$ is left-identity.

$$A \bullet B = B$$
$$B \bullet A = A$$

$Draw$ is non-commutative.

$$A \bullet B \neq B \bullet A$$

$Draw$ is associative.

$$(A \bullet B) \bullet C = (B) \bullet C = B \bullet C = C$$
$$A \bullet (B \bullet C) = A \bullet (C) = A \bullet C = C$$

$Draw$ is a primitive binary operator. There are four possible draws on pairs of primitives, and the interpretation is that the last primitive in a string of primitives is what is drawn.

$$F \bullet F = F$$
$$T \bullet F = F$$
$$F \bullet T = T$$
$$T \bullet T = T$$

The motivation is to introduce the smallest modification on $draw$ to be a logically complete Boolean algebra under $draw$. Before that, let's begin with a number of conventions.

Begin by no longer illustrating • so that AB is understood as $A \bullet B$.

A and B are $labels$ of the $value$ primitives.

Introduce $terms$ $P_\square$ and $Q_\square$ upon which a $draw$ is defined.

A $draws$ $P_\square$,

$$A \bullet P_\square = AP_\square = P_A = \begin{cases} P_F \\ P_T \end{cases}$$

where A,P_F and P_T are $labels$ of primitive $values$, and P_A is a $drawn$ $term$.

Two terms are equal when $concurrent$ $draws$ are the same. $P_\square$ and $Q_\square$ are equal when $AP_\square$ and $AQ_\square$ are the same, i.e. when P_F and Q_F are the same, and P_T and Q_T are the same then $P_\square$ = $Q_\square$.

Theorem 1.1. $P_\square A = A$

Proof. $P_\square$ is a $term$ and A is a primitive $value$. A has two cases: F and T.

(1) A = F: Assume $P_\square F$ then $P_\square$ $draws$ F which is F.

(2) A = T: Assume $P_\square T$ then $P_\square$ always $draws$ T by definition of $draw$.

$P_\square A$ is $concurrent$ with A, therefore, $P_\square A$ = A. □

P_A is a drawn $variable$ while P_F and P_T are $labels$ of binary $values$.

$$P_\square = \begin{cases} P_F = F \\ P_T = T \end{cases} \tag{1}$$

Under $AP_\square$, when $A = F$,

$$FP_\square = P_F = F.$$

and when $A = T$,

$$TP_\square = P_T = T.$$

Then, under (1), $AP_\square = A$.

$P_\square A = A$ because whatever $P_\square$ is, it draws A. The $term$ (1) under $draw$ now defines $right\text{-}identity$.

Theorem 1.2. $Identity$ *is* $P_\square$ = $\begin{cases} P_F = F \\ P_T = T \end{cases}$ *over* $AP_\square$.

Proof. For $P_\square$ to be $identity$, both $left\text{-}identity$ and $right\text{-}identity$ need to be satisified.

Consider first $AP_\square$. A is a primitive $value$ so there are two cases to consider, F and T.

(1) A = F: $FP_\square$ = P_F = F.

(2) A = T: $TP_\square$ = P_T = T.

A $draws$ on $P_\square$ and in both cases $AP_\square$ = A so $right\text{-}identity$ is shown.

Next, consider $P_{\square}A = A$. By theorem 1.1 and the definition of $draw$, this is the $left\text{-}identity$.

Both $left\text{-}identity$ and $right\text{-}identity$ are shown to be satisfied therefore $P_{\square}$ is the $identity$. $\square$

Let's label (1) as $identity\ \mathcal{I}_{\square}$. Note that $\mathcal{I}_{\square}$ is $commutative$ over $draw$ and $AP_{\square}$, i.e. $A\mathcal{I}_{\square} = \mathcal{I}_{\square}A$.

Next, let's define $P_{\square}$ as such:

$$P_{\square} = \begin{cases} P_F = T \\ P_T = F \end{cases} \tag{2}$$

When $A = F$,

$$FP_{\square} = P_F = T.$$

When $A = T$,

$$TP_{\square} = P_T = F.$$

Theorem 1.3. $Negation\ is\ P_{\square} = \begin{cases} P_F = T \\ P_T = F \end{cases}$ *over* $AP_{\square}$.

Proof. Consider $AP_{\square}$. A is a primitive $value$ so there are two cases to consider, F and T.

(1) $A = F$: $FP_{\square} = P_F$ = T.

(2) $A = T$: $TP_{\square} = P_T$ = F.

When A is $first$ the draw is $last$, and when A is $last$ then draw is $first$. Thus, $P_{\square}$ is $negation$. $\square$

The truth-term (2) defines the $negation$ of A so let's label (2) as $\mathcal{N}_{\square}$. Note that $\mathcal{N}_{\square}$ is not $commutative$, i.e. $A\mathcal{N}_{\square}$ is not necessarily equal to $\mathcal{N}_{\square}A$.

There are two more cases to consider.

$$P_{\square} = \begin{cases} P_F = F \\ P_T = F \end{cases} \tag{3}$$

$$P_{\square} = \begin{cases} P_F = T \\ P_T = T \end{cases} \tag{4}$$

Theorem 1.4. $Fallacy\ is\ P_{\square} = F = \begin{cases} P_F = F \\ P_T = F \end{cases}$ *over* $AP_{\square}$.

Proof. $P_\square$ is F when concurrent draws are the same.

A is a primitive $value$ so there are two cases to consider, F and T.

(1) $A = F$: $FP_\square = P_F = F$.

(2) $A = T$: $TP_\square = P_T = F$.

The $draw$ on $P_\square$ is F.

$AF = F$ by the definition of $draw$ and it follows two cases are to be considered, F and T.

(1) $A = F$: $FF = F$.

(2) $A = T$: $TF = F$.

$FP_\square = F = FF$ and $TP_\square = F = TF$.

The concurrent $draws$ are the same therefore $P_\square = F$. □

Corollary 1.4.1. $Tautology$ *is* $P_\square = T = \begin{cases} P_F = T \\ P_T = T \end{cases}$ *over* $AP_\square$.

These two special cases can be considered as $false$ and $true$. Let (3) be the truth-term $\mathcal{F}_\square\ draw\ false$, and (4) be $\mathcal{T}_\square\ draw\ true$.

The cases (3) and (4) can be generalized. The $truth\text{-}values$ can be combined into the special case (5) which is then just $left\text{-}identity \bullet right\text{-}draw$.

$$BP_\square = P_B = \begin{cases} P_F = B \\ P_T = B \end{cases} \tag{5}$$

$$AP_B = B$$
$$P_BA = A$$

And now, $AP_\square$ extends $draw$, $A \bullet B$, along with $identity\ \mathcal{I}_\square$, $negation\ \mathcal{N}_\square$, $false\ \mathcal{F}_\square$, and $true\ \mathcal{T}_\square$.

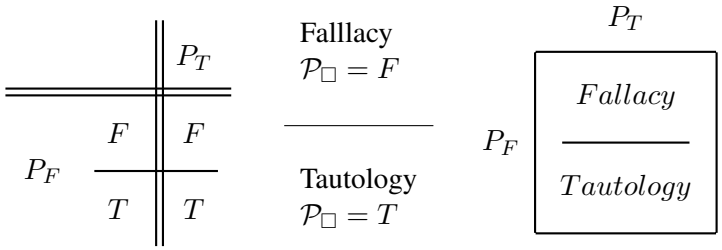


Figure 5: $Fallacy$ and $Tautology$

Starting with the Cayley table (Figure 5, left), draw P_F. Map each draw to a unique position in the Boolean group table (Figure 5, right). Show that each choice in the Boolean group table maps to Cayley table *one-to-one* and *onto*. This shows the equivalence of tables. Show with theorem 1.4 and corollary 1.4.1 that the Boolean $grid$ (Figure 5, middle) is isomorphic to the Cayley table and Boolean group table.

		P_T	
		F	T
P_F	F	False	Identity
	T	Negation	True

Figure 6: Cayley Table

False $\mathcal{F}_\square = \begin{cases} \mathcal{F}_F = F \\ \mathcal{F}_T = F \end{cases}$	Identity $\mathcal{I}_\square = \begin{cases} \mathcal{I}_F = F \\ \mathcal{I}_T = T \end{cases}$
Negation $\mathcal{N}_\square = \begin{cases} \mathcal{N}_F = T \\ \mathcal{N}_T = F \end{cases}$	True $\mathcal{T}_\square = \begin{cases} \mathcal{T}_F = T \\ \mathcal{T}_T = T \end{cases}$

Figure 7: Inverted Cayley Table

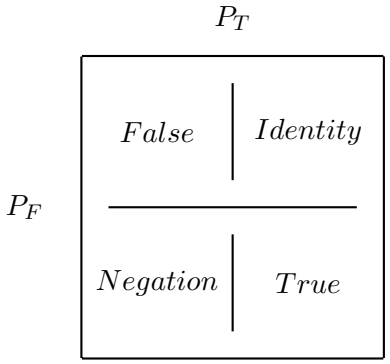


Figure 8: Cayley Table Isomorphism

False $\mathcal{F}_{\square} = F$	Identity $\mathcal{I}_{\square} = \begin{cases} \mathcal{I}_F = F \\ \mathcal{I}_T = T \end{cases}$
Negation $\mathcal{N}_{\square} = \begin{cases} \mathcal{N}_F = T \\ \mathcal{N}_T = F \end{cases}$	True $\mathcal{T}_{\square} = T$

Introducing a notation

$$\{P_{\square}, P_F, P_T\} \tag{6}$$

that is the same as

$$P_{\square} = \begin{cases} P_F \\ P_T \end{cases}$$

.

Example, $\mathcal{I}_{\square} \equiv \{P_{\square}$, P_F = F, P_T = T$\}$.

Let's consider two truth-values of another special case. The above definitions (1)-(5) are drawn of single truth-values so define a new truth-term drawn of two truth-values of the following form.**

$$B(AP_{\square})$$

P_F \ P_T	$\mathcal{F}_{\square}$	$\mathcal{I}_{\square}$	$\mathcal{N}_{\square}$	$\mathcal{T}_{\square}$
$\mathcal{F}_{\square}$	False	And	$\nrightarrow$	Is
$\mathcal{I}_{\square}$	$\nleftarrow$	Identity	Xor	Or
$\mathcal{N}_{\square}$	Nor	Xnor	Negation	$\leftarrow$
$\mathcal{T}_{\square}$	Is Not	$\rightarrow$	Nand	True

Figure 9: Cayley Table

Theorem 1.5. *$\{\mathcal{I}_{\square}$, $\mathcal{I}_F$ = F, $\mathcal{I}_T$ = T$\}$ over $B\mathcal{I}_{\square}$ is concurrent with $\{P_{\square}$, P_F = $\mathcal{I}_{\square}$, P_T = $\mathcal{I}_{\square}\}$ over $B(AP_{\square})$.*

Proof. $P_{\square}$ is $\mathcal{I}_{\square}$ when concurrent draws are the same.

Firstly, $P_{\square}$ over $B(AP_{\square})$ is easier to consider when A draws $P_{\square}$, it is $\mathcal{I}_{\square}$ so $B(AP_{\square})$ = $B(\mathcal{I}_{\square})$ = $B\mathcal{I}_{\square}$ = $\mathcal{I}_B$.

False $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	And $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	$\nrightarrow$ $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	Is $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
$\nleftarrow$ $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	Identity $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	Xor $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	Or $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
Nor $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	Xnor $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	Negation $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	$\leftarrow$ $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
Is Not $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	$\rightarrow$ $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	Nand $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	True $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{T}_\square \end{cases}$

Figure 10: Cayley Table $P_F \times P_T$

	F•P_F	F•P_T
T•P_F	$False$ / And	$\nrightarrow$ / Is
T•P_T	Nor / $Xnor$	$Negation$ / $\leftarrow$
T•P_F	$\nleftarrow$ / $Identity$	Xor / Or
T•P_T	$IsNot$ / $\rightarrow$	$Nand$ / $True$

Figure 11: Boolean Group

There are two concurrent draws to consider on B.

(1) B = F: $\mathcal{I}_F$ = F.

(2) B = T: $\mathcal{I}_T$ = T.

The $draw$ on $P_\square$ is B.

The $draw$ on $\mathcal{I}_\square$ have two cases for B:

(1) B = F: $F\mathcal{I}_\square$ = $\mathcal{I}_F$ = F.

(2) B = T: $T\mathcal{I}_\square$ = $\mathcal{I}_F$ = T.

$\{\mathcal{I}_\square, \mathcal{I}_F = F, \mathcal{I}_T = T\}$ is $concurrent$ with $\{P_\square, P_F = \mathcal{I}_\square, P_T = \mathcal{I}_\square\}$. □

Corollary 1.5.1. $P_\square = \mathcal{N}_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{N}_\square \end{cases}$ *over* $AP_\square$.

False $P_\square = F$	And $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	$\nrightarrow$ $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	Is $P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
$\nleftarrow$ $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	Identity $P_\square = \mathcal{I}_\square$	Xor $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	Or $P_\square = \begin{cases} P_F = \mathcal{I}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
Nor $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	Xnor $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	Negation $P_\square = \mathcal{N}_\square$	$\leftarrow$ $P_\square = \begin{cases} P_F = \mathcal{N}_\square \\ P_T = \mathcal{T}_\square \end{cases}$
Is Not $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{F}_\square \end{cases}$	$\rightarrow$ $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{I}_\square \end{cases}$	Nand $P_\square = \begin{cases} P_F = \mathcal{T}_\square \\ P_T = \mathcal{N}_\square \end{cases}$	True $P_\square = T$

$$P_\square = \begin{cases} P_F = \mathcal{F}_\square \\ P_T = \mathcal{I}_\square \end{cases} \tag{7}$$

Let's define (7) as And. There are four cases to consider for A and B drawing on B(A$P_\square$) where $P_\square$ = ($F(\mathcal{F}_\square)T(\mathcal{I}_\square)$) (*see* figure (12))

It would be helpful to take the form BA$P_\square$ and describe the isomorphic table. The motivation is to describe the table configuration from the algebraic notation. A new set {H, V, O} serves as a starting point.

Even though $P_\square$ can be P_H or P_V these are not draws. $B\bullet P_\square$, for example, is still a draw on the term by the truth-variable but relative to the orientation of the term, $B\bullet P_H$, or $B\bullet P_V$, in this example. $P_\square$ is $assigned$ to either binary orientations P_H or P_V.

AB			
FF	FT	TF	TT
$F(FP_{\square})$	$T(FP_{\square})$	$F(TP_{\square})$	$T(TP_{\square})$
$F(P_F)$	$T(P_F)$	$F(P_T)$	$T(P_T)$
$F(\mathcal{F}_{\square})$	$T(\mathcal{F}_{\square})$	$F(\mathcal{I}_{\square})$	$T(\mathcal{I}_{\square})$
$F\mathcal{F}_{\square}$	$T\mathcal{F}_{\square}$	$F\mathcal{I}_{\square}$	$T\mathcal{I}_{\square}$
$\mathcal{F}_F$	$\mathcal{F}_T$	$\mathcal{I}_F$	$\mathcal{I}_T$
F	F	F	T

Figure 12: Definition of And over BA•($F(\mathcal{F}_{\square})T(\mathcal{I}_{\square})$)

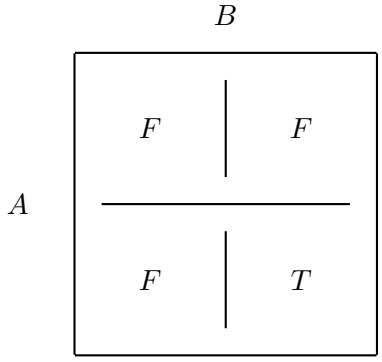


Figure 13: $Truth\text{-}table$ of And over $B(AP_{\square})$ where $\{P_{\square}, P_F = \mathcal{F}_{\square}, P_T = \mathcal{I}_{\square}\}$

In the following example,

$$\{P_{\square}, P_F = \mathcal{F}'_{\square}, P_T = \mathcal{I}'_{\square}\}$$

indicates either

$$\{P_H, P_F = \mathcal{F}_V, P_T = \mathcal{I}_V\}$$
$$\{P_V, P_F = \mathcal{F}_H, P_T = \mathcal{I}_H\}$$

B(A$P'_{\square}$) where $\{P'_{\square}, P_F = \mathcal{F}_{\square}, P_T = \mathcal{I}_{\square}\}$ is equivalent to either of the following:

$$HVOOVOO, FFFT$$

$$VHOOHOO, FFFT$$

$$\{P_H, P_F = \{\mathcal{F}_V, \mathcal{F}_F = F, \mathcal{F}_T = F\}, P_T = \{\mathcal{I}_V, \mathcal{I}_F = F, \mathcal{I}_T = T\}\}$$

$$\{P_H, P_F, P_T\} \equiv HOO, P_F P_T$$

$$\{P_V, P_F, P_T\} \equiv VOO, P_F P_T$$

The following definitions can be mapped, arbitrarily to H or V:

$$False \equiv \{P_V, P_F = F, P_T = F\}$$
$$VOO, FF$$

$$Identity \equiv \{P_H, P_F = F, P_T = T\}$$
$$HOO, FT$$

It will be advantageous to $recursively$ alternate between H and V when assigning orientations in that $P_\square \bullet (HV)$ is the desired permutation to preserve $symmetry$.

$$And \equiv \{P_H, P_F = \mathcal{F}_V, P_T = \mathcal{I}_V\}$$
$$HOO, \mathcal{F}_\square \mathcal{I}_\square$$

$P_\square$ is assigned H or V orientation but will still be a draw on P_F and P_T. $P_\square$ maps to H or V. When the draw is F or T, P_F or P_T is assigned to O but if the draw is some subsequent term $Q_\square$ then Q_F or Q_T is assigned to either H or V. This $convention$ continues rescursively.

$$And \equiv \{P_H, P_F = \{\mathcal{F}_V, \mathcal{F}_F = F, \mathcal{F}_T = F\}, P_T = \{\mathcal{I}_V, \mathcal{I}_F = F, \mathcal{I}_T = T\}\}$$
$$HVOOVOO, FFFT$$

By theorem 1.4 and symmetry, the preceding definition of And (see figure 13) is equivalent to the following (also see figure 14).

$$And \equiv \{P_V, P_F = F, P_T = \{\mathcal{I}_H, \mathcal{I}_F = F, \mathcal{I}_T = T\}\}$$
$$VOHOO, FFT$$

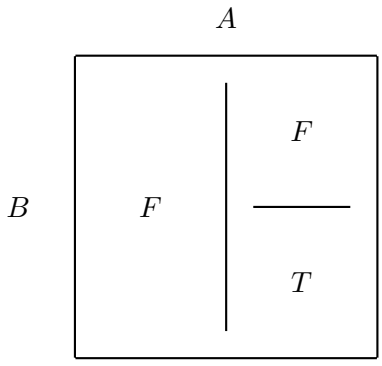


Figure 14: $And \equiv B(AP_V)$ where $\{P_V$, P_F = F, P_T = $\{\mathcal{I}_H, \mathcal{I}_F = F, \mathcal{I}_T = T\}\}$

Note that figure 14 is described resursively by $VOHOO, FFT$ and each O is corresponds with truth-values.

2 Summary

From the primitives of $fallacy$ and $tautology$ the first $unary$ operators are defined. Together, the $primitives$ and $unary$ operators define the $binary$ operators. $Logical\ algebraic\ proofs$ of the $premises$ are $deduced$. Finally, the $logic$ is represented by $binary$ diagrams in $relation$ to the algebra.